GROUP 8

DATA GOVERNANCE TASK FORCE HIRED BY NOVACRED

NOVACRED

INTELLIGENT CREDIT DECISIONS

Data · Governance · Fairness

TEAM

CREDIT APLLICATION GOVERNANCE AUDIT

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PRODUCT LEAD

PROBLEM

NovaCred

A fintech company using machine learning to approve loans, has received a regulatory inquiry about possible discrimination in its lending decisions.

The company must investigate whether its credit decision system is ***fair, legally compliant****, and* ***built on reliable data****.*

Data Quality Risk

The credit application dataset may contain:

- missing or inconsistent values
- incorrect data types
- invalid financial information
- Poor data quality can lead to incorrect model decisions.

Algorithm Bias Risk

Loan approvals may unintentionally discriminate:

- Gender
- Age
- geographic proxies (e.g., ZIP code)

If bias exists, the system could violate fairness standards

Privacy & Governance Risk

The dataset contains sensitive personal data:

- full name
- Email
- SSN
- IP address
- date of birth

If not properly managed, this may violate GDPR and EU AI Act requirements.

DATASET OVERVIEW

RAW CREDIT APPLICATION RECORDS FROM NOVACRED

502 records | nested JSON | 4 categories | Mongo DB collection

Applicant Info →
- Full Name
- Email
- **SSN**
- Gender
- Date of Birth
- ZIP Code
- IP Address

Financial Data →
- Annual Income
- Debt-to-Income Ratio
- Credit History (mo.)
- Savings Balance

Spending Behavior →
- Category
- Amount
- (array of objects)

Loan Decision →
- Loan Approved
- Approved Amount
- Interest Rate
- Rejection Reason

** loan_purpose and processing_timestamp present in subset of records*

= Sensitive / PII field

DATA QUALITY ISSUES

Issue Type	Dimension	Affected	Risk	Fix
Duplicate Records	Accuracy	4 (0.8%)	Inflates totals, distorts approval rates	Drop duplicates by _id, keep first
Missing Values	Completeness	5 (1.0%)	Unusable records, potential bias	Flag via has_critical_missing boolean
Invalid Values	Validity	13 (2.6%)	Corrupts downstream analysis	Set impossible values to NaN
Inconsistent Gender Coding	Consistency	113 (22.6%)	Splits same group into separate buckets	Map all variants to Male/Female
Mixed Date Formats	Consistency	161 (32.2%)	Wrong ages, silent data loss	Parse all to single datetime column
Out-of-Range Numerics	Validity	3 (0.6%)	Corrupts averages and model training	NaN for impossible; flag outliers
Nested spending_behavior	Consistency	500 (100%)	Unusable for analysis or CSV export	Pivot to one column per category
Redundant Income Field	Consistency	5 (1.0%)	5 records silently excluded from stats	Merge salary into income, drop column
Wrong Data Types	Consistency	34 of 35 cols	Aggregation failures, bad filtering	Coerce all columns to expected types

BIAS RESULTS

FAIRNESS IN LOAN APPROVAL DECISIONS

DI = 0.77	66 %	51 %	-15 pp
Disparate Impact – threshold < 0.80	Male approval rate	Female approval rate	Approval gap (SPD)

Gender Bias · Flagged by the 4/5ths Rule

- DI = 0.77, below the 0.80 threshold signals potential discrimination
- Male applicants have ~88% higher odds of approval (logistic regression, $p < 0.001$)
- Financial profiles (income, savings, debt-to-income) are similar across genders, gap is not justified by data

Age × Gender Interaction

- Younger applicants (18–29) face the lowest approval rates, consistent with industry norms
- Gender gap persists across all age groups: 18–29, 30–44, and 45–59
- Gap narrows only in the 60+ group, where disparity is minimal

ZIP Code as a Proxy for Gender

- High-approval ZIP codes (≥ 80% rate): 68% male applicants
- Low-approval ZIP codes (≤ 33% rate): 70% female applicants
- ZIP code may act as an indirect gender proxy, embedding bias without using gender explicitly

Spending Patterns · Gambling as a Hidden Proxy

- Gambling spending recorded exclusively among female applicants in this dataset
- Models typically penalise gambling heavily as a risk signal
- This creates an indirect channel that disadvantages female applicants

PRIVACY RISKS

PII Fields

Full Name, Email, SSN, Gender, Birthday, ZIP-Code, IP-Address

Identifier

Full Name: Steffen Tigges

Hash

Salt

Pseudonymization of PII Fields

Full Name: ube8hdqwu938djwe9

Quasi-Identifier

ZIP-Code: 50825

k-1

Anonymize

Anonymization of PII Fields

ZIP-Code: 508**

GOVERNANCE SOLUTIONS

GDPR Art. 5: Data Minimization	Delete fields Adult Entertainment and Healthcare
GDPR Art. 5.1a: Fairness	Remove fields Gender and Gambling from the Algorithm to prevent Discrimination
GDPR Art. 5.1: Retention Policy	Delete entries after 7 years, IP Address after 90 days
GDPR Art. 17: Right to be forgotten	Delete personal data upon request, but keep decision data
GDPR Art.22 & EU AI Act Art.14: Human i. t. Loop	Add fields: AI Recommendation and AI Reasoning, but Human Operator makes final decision
GDPR Art. 30 & EU AI Act Art. 12: Audit Trail	Create new DF: Log every time the data is accessed or altered by operators or the system

CONCLUSIONS

NOVACRED GOVERNANCE AUDIT

Data quality issues found at intake:
Raw dataset contained duplicates, **12%** unrecognised date formats, inconsistent gender encoding, and invalid financial values, all remediated in the cleaning pipeline

Gender bias confirmed:
Female applicants face a **51%** approval rate vs. **66%** for males, a Disparate Impact ratio of **0.77**, below the legal **80%** threshold, with male applicants showing **~88%** higher odds of approval (significant at $p < 0.001$)

Proxy discrimination detected:
Low-approval ZIP codes are **70%** female; gambling spending is exclusive to female applicants and likely penalized by the model, indirect discrimination channels exist

Serious GDPR gaps identified:
Handling of sensitive data, no data retention policy, no audit trail, and no human oversight, all four represent **violations under GDPR** Articles 5, 6, 9,17, 22, and 30

EU AI Act classifies this as High-Risk:
Credit scoring falls under strict obligations, requiring bias-free training data, full documentation, and mandatory human-in-the-loop oversight

Governance controls implemented:
Pseudonymization of PII, consent tracking **(95% opt-in),** retention policy (5/10-year tiers), right-to-erasure workflow, audit logging, and Human-in-the-Loop architecture all deployed

Do not deploy the current model as-is.

The AI should function strictly as an advisory tool under a Human-in-the-Loop architecture. Before any production deployment, conduct a full fairness audit, remove proxy variables (ZIP code, gambling spending), and complete GDPR/AI Act documentation.